

Filip Bělík

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PhD Candidate

Department of Mathematics and Scientific Computing and Imaging Institute
University of Utah, Salt Lake City, USA

EDUCATION

• University of Utah

Mathematics PhD: Applied & Computational Mathematics

August 2022 - Present

Salt Lake City, UT, USA

- Co-advised by [Dr. Akil Narayan](#) and [Dr. Christel Hohenegger](#)
- Intended graduation in May 2027
- President of University's [Student SIAM chapter](#)
- Chair of Mathematics [Graduate Student Recruitment Committee](#)
- GPA: 4.00

Coursework:

- MATH 5080 Statistical Inference I
- MATH 6010 Linear Models
- MATH 6210 Real Analysis
- MATH 6410 Ordinary Differential Equations
- MATH 6420 Partial Differential Equations
- MATH 6610 Analysis of Numerical Methods I
- MATH 6620 Analysis of Numerical Methods II
- MATH 6630 Numerical Method for Partial Differential Equations
- MATH 6710 Applied Linear Operators and Spectral Methods
- MATH 6720 Applied Complex Variables and Asymptotic Methods
- MATH 6740 Bifurcation Theory
- MATH 6750 Fluid Dynamics
- MATH 6880 Mathematics of Data Science
- MATH 7875 Advanced Optimization

• Gustavus Adolphus College

BA Honors Mathematics & BA Computer Science

September 2018 - May 2022

St. Peter, MN, USA

- Student host for [Nobel Conference 2021, Big Data](#)
- Mathematics, Computer Science, and Statistics (MCS) Department Assistant, 2021
- President of Club Tennis; Co-President of Coding Club; MCS Club; Running Club
- Cumulative GPA: 3.989
- Major GPA: 4.00

Coursework:

- MCS-150 Discrete Mathematics
- MCS-177 Computer Science I (Python)
- MCS-178 Computer Science II (Java/Kotlin/Assembly)
- MCS-220 Introduction to Analysis
- MCS-221 Linear Algebra
- MCS-222 Multivariable Calculus
- MCS-256 Discrete Calculus
- MCS-265 Theory of Computation
- MCS-270 Android Development
- MCS-284 Computer Organization (C)
- MCS-313/314 Modern Algebra I and II
- MCS-321 Theory of Complex Variables
- MCS-331 Real Analysis
- MCS-353 Continuous Dynamical Systems
- MCS-355 Scientific Computing
- MCS-357 Discrete Dynamical Systems
- MCS-375 Algorithms
- MCS-377 Networking

• East Ridge High School

Secondary Education

September 2015 - May 2018

Woodbury, MN, USA

- Varsity Tennis
- Mathematics Club
- Weighted GPA: 4.123
- Unweighted GPA: 3.814

WORK EXPERIENCE

• University of Utah Mathematics Department

Graduate Research and Teaching Assistant

August 2022 - Present

Salt Lake City, UT, USA

- Instructor of record for MATH 1310, Engineering Calculus I, Spring 2025
- Research funding under Dr. Narayan, Dr. Christel Hohenegger, and a University of Utah funding incentive seed grant
- Lab TA for MATH 4600 Mathematics in Medicine

• Gustavus Mathematics and Computer Science Department

TA & Tutor & Grader

February 2019 - May 2022

St. Peter, MN, USA

- Computer Science I Teacher's Assistant, Grader, and Tutor (Python)
- Computer Science II Teacher's Assistant and Tutoring (Kotlin and Java)
- Discrete Mathematics Grader
- Online volunteer tutoring during COVID semester

• Allianz Life

Hedging Intern

May 2021 - August 2021

Golden Valley, MN, USA

- Software development with C# and SQL
- Implemented procedure for automating the labeling of incoming market data
- Developed application for visualization of 3D data and interpolation and approximation of data
- Presented work to market data team

• Her Next Play

CRM Intern

June 2020 - August 2020

Remote

- Research and implementation of contact resource management software
- Presentation and instruction to executives

RESEARCH

• Model Order Reduction and Numerical Methods for Conservation Laws

with Dr. Akil Narayan

May 2023 - Present

- Study and implementation of finite difference, finite volume, discontinuous Galerkin methods for conservation laws
- Study and implementation of parametric model order reduction methods for linear stationary and nonstationary parametric problems
- Goal of expanding to nonlinear and transport dominated problems
- Development of open-source software package [ModelOrderReductionToolkit.jl](#)

• Blood Flow Modeling for Conductivity and Uncertainty Quantification

with Dr. Christel Hohenegger

June 2022 - Present

- Analytical modeling of arterial blood flow in arterial tree and resulting electrical properties
- Goal of better understanding relationship between blood pressure and blood-driven electrical properties at the wrist
- Work to understand impacts of physiological parameters through local and (Sobol-based) global sensitivity analyses
- Collaboration with [Henry Crandall](#) and [Dr. Benjamin Sanchez](#) (University of Utah Electrical Engineering) and Tyler Schuessler and [Dr. Braxton Osting](#) (University of Utah Mathematics)

• Carathéodory-Steinitz Pruning

with Dr. Akil Narayan and Dr. Jesse Chan (Rice University)

May 2023 - Present

- Algorithm design and stability analysis for constructing positive, efficient, nested quadrature rules
- Testing of various methods for numerical quadrature and model order reduction applications
- Development of open-source Julia package [CaratheodoryPruning.jl](#)

• Uncertainty Quantification for Markov Decision Processes in Ecological Settings

with John Turnage, Dr. Akil Narayan and Dr. Jody Reimer

May 2023 - Present

- Study of use of decision-based models for ecological processes

- Identify uncertainty quantification methods for Markov decision processes
- **Modeling Closed Vortices as Self-Avoiding Polygons** August 2021 - May 2022
with *Dr. Pavel Bělík* (Augsburg University) and *Dr. Thomas LoFaro* (Gustavus Adolphus College)
 - Undergraduate mathematics honors project
 - Modeled closed-loop vortices, such as dolphin bubble rings or smoke rings, as self-avoiding polygons in the cubic lattice
 - Implemented sets of transformations for use in Metropolis Markov Chain Monte-Carlo methods
- **Port-and-Sweep Solitaire Army Problem** May 2020 - September 2020
with *Dr. Jacob Siehler* (Gustavus Adolphus College)
 - Studied mathematics of [Port-and-Sweep Solitaire](#)
 - Used algebraic and computational techniques to solve one-dimensional army problem
- **The Propagation of Health-Related Habits on Twitter** May 2019 - November 2019
with *Dr. Louis Yu* and *Jeffery Engelhardt* (Gustavus Adolphus College)
 - Accepted as one of six first-year Gustavus students for ten-week research project under First-Year Research Experience (FYRE)
 - Use of various machine learning models in classification of tweets
 - Construction of listener to run over twelve-week period
 - Presented and attended research presentations at Midstates Consortium at University of Chicago

PUBLICATIONS/PREPRINTS

1. Filip Bělík, Yanlai Chen, and Akil Narayan. *Greedy Rational Approximation for Frequency-Domain Model Reduction of Parametric LTI Systems*. 2025. arXiv: [2512.23814](#) [math.NA]. URL: <https://arxiv.org/abs/2512.23814>
2. Filip Bělík, Jesse Chan, and Akil Narayan. *Efficient and Robust Carathéodory-Steinitz Pruning of Positive Discrete Measures*. 2025. arXiv: [2510.14916](#) [math.NA]. URL: <https://arxiv.org/abs/2510.14916>

CONFERENCES

- **Joint Mathematics Meetings** January 2026
Walter E. Washington Convention Center, Washington D.C. 
 - Talk: Greedy Rational Approximation for Model Reduction of LTI Systems
- **Wasatch SIAM Student Chapters Conference** April 2025
Utah State University, Logan UT 
 - Talk: Carathéodory-Steinitz Pruning for Numerical Quadrature
- **MAA North Central Section Meeting** March 2025
St. Olaf College, Northfield MN 
 - Talk: Carathéodory-Steinitz Pruning for Numerical Quadrature
- **Model Reduction and Surrogate Modeling** September 2024
Scripps Seaside Forum, La Jolla CA 
 - Talk: Greedy Frequency Domain Model Reduction for Parametric Systems: New Theory and Algorithms
- **NSF Computational Mathematics PI Meeting** July 2024
University of Washington, Seattle WA 
 - Poster: Dynamic Bulk Conductivity in Radial Artery
- **Mathematical Opportunities in Digital Twins** December 2023
George Mason University, Arlington VA 
 - Poster: Dynamic Bulk Conductivity in Radial Artery
- **Midstates Consortium for Math and Science** November 2019
University of Chicago, Chicago IL 
 - Talk: The Propagation of Health-Related Habits on Twitter

HONORS AND AWARDS

- **Outstanding Graduate Student** May 2025
University of Utah Mathematics Department
- **RTG: Optimization and Inversion Summer Grant** May 2023
University of Utah Mathematics Department 
- **Fulbright Canada Mitacs Globalink Research Internship** April 2021
Mitacs Globalink 
- **Hilding Research Fund** April 2020
Gustavus Adolphus College
- **First-Year Research Experience (FYRE)** April 2019
Gustavus Adolphus College 
- **Math Problem Solving Competition** 2018, 2019
Gustavus Adolphus College 
- **Dean's Scholarship** Fall 2018
Gustavus Adolphus College

SKILLS

- **Programming Languages:** Julia, Python, MATLAB, LaTeX, C, C++, Java, C#
- **Computer Operating Systems:** Windows, Linux, Mac
- **Web Development:** HTML, JavaScript, CSS
- **Mathematical/Technical Tools:** Jupyter, LaTeX, Maple, SQL
- **Android App Development:** Kotlin
- **Competitive Programming:** ICPC, COMAP, Kattis (fbelik), Project Euler (fbelik)